Compiler Construction

Lab 6

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53953

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Task 1:

```
%option noyywrap
%{
#include <stdio.h>
#include <string.h>

int keyword_count = 0;
int identifier_count = 0;
int operator_count = 0;
int numeric_count = 0;

// List of C keywords
char *keywords[] = {
    "auto","break","case","char","const","continue","default","do","double",
    "else","enum","extern","float","for","goto","if","int","long","register",
    "return","short","signed","sizeof","static","struct","switch","typedef",
    "union","unsigned","void","volatile","while", NULL
};

// Check if a word is a keyword
int is_keyword(char *word) {
    for (int i = 0; keywords[i] != NULL; i++) {
        if (strcmp(word, keywords[i]) == 0)
            return 1;
    }
    return 0;
}
%}
```

```
%%

[ \t\n]+                ;   // Ignore whitespace

"=="|"<="|">="|"!="|"="|"+"|"-"|"*"|"/"|"%"|"<"|">" {
    operator_count++;
    printf("OPERATOR: %s\n", yytext);
}

[0-9]+\.[0-9]+          { numeric_count++; printf("NUMBER: %s\n", yytext); }  // Float literals
[0-9]+                  { numeric_count++; printf("NUMBER: %s\n", yytext); }  // Integer literals

[a-zA-Z_][a-zA-Z0-9_]*  {
    if (is_keyword(yytext)) {
        keyword_count++;
        printf("KEYWORD: %s\n", yytext);
    } else {
        identifier_count++;
        printf("IDENTIFIER: %s\n", yytext);
    }
}

"(" | ")" | "{" | "}" | ";" | ","  { printf("PUNCTUATION: %s\n", yytext); }

.                       ; // Ignore other characters

%%
```

```
int main(int argc, char **argv) {
        printf("Syeda Ayesha Fatima\n");
    printf("Enter code (Ctrl+Z then Enter to stop):\n");
    yylex(); // Start scanning

    printf("\nSummary:\n");
    printf("Keywords: %d\n", keyword_count);
    printf("Identifiers: %d\n", identifier_count);
    printf("Operators: %d\n", operator_count);
    printf("Numeric literals: %d\n", numeric_count);

    return 0;
}
```

```
D:\Desktop\Lex Analyzer>flex CC-L6-1.l

D:\Desktop\Lex Analyzer>gcc lex.yy.c

D:\Desktop\Lex Analyzer>a.exe
Syeda Ayesha Fatima
Enter code (Ctrl+Z then Enter to stop):
string name="Ayesha";
IDENTIFIER: string
IDENTIFIER: name
OPERATOR: =
IDENTIFIER: Ayesha
int age=20;
KEYWORD: int
IDENTIFIER: age
OPERATOR: =
NUMBER: 20
^Z

Summary:
Keywords: 1
Identifiers: 4
Operators: 2
Numeric literals: 1
```

Task 2:

```
%option noyywrap
%{
#include <stdio.h>

int line = 1;
int start = 1;   // to print line number once per line
%}

%option noyywrap
%x COMMENT

%%

"//".*          ;                   /* remove single line comment */

"/*"            { BEGIN(COMMENT); }     /* start multi-line comment */

<COMMENT>[^*\n]*    ;                   /* ignore text */

<COMMENT>"*"+[^*/\n]* ;

<COMMENT>\n         ;                   /* ignore newline in comment */

<COMMENT>"*"+"/"    { BEGIN(INITIAL); }     /* end comment */

\n              {
                  printf("\n");
```

```
                line++;
                start = 1;
            }

.           {
                if(start){
                    printf("Line %d: ", line);
                    start = 0;
                }
                printf("%s", yytext);
            }
%%
int main()
{
        printf("Syeda Ayesha Fatima\n");
        printf("Enter code (Ctrl+Z then Enter to stop):\n");
    yylex();
    return 0;
}
```

```
D:\Desktop\Lex Analyzer>flex CC-L6-2.l

D:\Desktop\Lex Analyzer>gcc lex.yy.c

D:\Desktop\Lex Analyzer>a.exe
Syeda Ayesha Fatima
Enter code (Ctrl+Z then Enter to stop):
string name="Ayesha";
Line 1: string name="Ayesha";
if (name=="Ayesha");
Line 2: if (name=="Ayesha");
printf("Present");
Line 3: printf("Present");
^Z
```

Task 3:

```
%option noyywrap
%{
#include <stdio.h>
%}
%option noyywrap
%%
"++"|"--"|"="|"+="|"-="|"*="|"/="|"%="    { printf("Assignment Operator: %s\n", yytext); }
"<="|">="|"=="|"!="|"<"|">"               { printf("Relational Operator: %s\n", yytext); }
"+"|"-"|"*"|"/"|"%"                       { printf("Arithmetic Operator: %s\n", yytext); }
[ \t\n]+                                  ;   /* ignore whitespace */
.                                         ;   /* ignore other characters */
%%
int main()
{
        printf("Syeda Ayesha Fatima\n");
   printf("Enter expression (Ctrl+Z then Enter to stop):\n");
   yylex();
   return 0;
}
```

```
D:\Desktop\Lex Analyzer>flex CC-L6-3.l

D:\Desktop\Lex Analyzer>gcc lex.yy.c

D:\Desktop\Lex Analyzer>a.exe
Syeda Ayesha Fatima
Enter expression (Ctrl+Z then Enter to stop):
int a=10;
Assignment Operator: =
a<=10
Relational Operator: <=
a=a+5
Assignment Operator: =
Arithmetic Operator: +
^Z
```